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Portugal

TECHNICAL SKILLS

PROGRAMMING

Python · JavaScript · SQL
Go (project experience)

BACKEND & DATA

FastAPI · Node.js
PostgreSQL · Redis · SQLite
Supabase · DuckDB · dbt

INFRASTRUCTURE

Docker · Kubernetes · Terraform
GCP · Prometheus
(project deployments)

LANGUAGES

Spanish - Native
Portuguese - Fluent
English - Advanced

HOW I WORK

AI-assisted implementation.
I make architectural decisions,
review code and test behaviour.

WORK AVAILABILITY

Portugal: no sponsorship needed.
Contractor / B2B available.

PROJECT LINKS

Nexus - live research system
FOAAP
GitHub - source code

PAULO VILLALOBOS

SOFTWARE DEVELOPER
BACKEND SYSTEMS ENGINEERING

I build APIs, data pipelines and background services, with a focus on data integrity, recoverable processing and inspectable state. My projects cover PostgreSQL versioning, Go workers and evidence-governance systems.

Remote · Full-time or part-time · Available immediately

SELECTED ENGINEERING PROJECTS

Independent projects | 2026

AUDIO JOBS | Background processing

Go · PostgreSQL · Redis Streams

Built an asynchronous audio-processing service with bounded worker concurrency, lease-fenced writes, retries and recovery of jobs lost from the queue. A recorded local failure-injection run completed **1,000/1,000 jobs**.

Code and tests

MACRO VINTAGES | Versioned data API

Python · FastAPI · PostgreSQL

Built ingestion and an API that retain changes to economic data and answer point-in-time queries. Used transactional writes, non-overlapping version intervals, concurrent job claiming, API keys and rate limits.

Code and tests

NEXUS / MESH | Evidence governance

Python · SQLite · Research information systems

Built revision history, source validation and independent review gates for evidence-linked research. Transactional commands reject stale edits; publication freezes a verifiable snapshot. **42 governance tests pass.**

Code and tests

EDUCATION

Capella University
Bachelor's studies in computer science - in progress

SELECTED WORK

APIs / Data integrity / Background processing / Infrastructure

01 / AUDIO JOBS + PLATFORM INFRA

Go · PostgreSQL · Redis · Kubernetes · Terraform · GCP

Built the processing service and its infrastructure as independent engineering projects. PostgreSQL holds job state; Redis Streams delivers work. Workers acquire leases and fence result writes so an expired attempt cannot overwrite a newer one.

SUBMIT → PERSIST → QUEUE → LEASE → PROCESS → COMMIT

Added retry backoff, dead-letter handling, stale-job recovery, graceful shutdown, metrics and failure drills. Infrastructure code includes a private GKE cluster, workload identity, network policies, health probes, resource limits and deployment controls.

Recorded results: a local synthetic benchmark increased throughput from **161.5 to 532.3 jobs/s** with streaming analysis. A separate local chaos run completed **1,000/1,000 jobs** despite worker kills and a Redis restart. In a temporary GCP node-loss drill, **1,800 accepted jobs completed**. These are project test results, not customer production metrics.

Service + tests / Benchmark record / GCP validation

02 / MACRO VINTAGES

Python · FastAPI · PostgreSQL

Preserves each economic observation as the engine learns revisions. Database constraints prevent overlapping version intervals; source values and write history remain queryable.

Workers claim jobs with **FOR UPDATE SKIP LOCKED** and commit only while holding the current lease. The API adds hashed keys, a database-backed token bucket, cursor pagination and ETags.

Point-in-time history reflects what the engine observed, not reconstructed publisher vintages.

API, schema and race tests

03 / NEXUS + MESH

Python · SQLite · Evidence lineage

Nexus makes research information explorable. Mesh provides the governance layer: immutable revisions, exact source references, independent reviews and controlled publication.

Commands run transactionally with actor-scoped idempotency. Changed sources invalidate dependent assertions; reviewed publication requires approval of the exact current revision.

42 automated checks pass locally, covering rollback, concurrency, permissions and evidence integrity.

Open Nexus / Code + tests

ALSO BUILT / FOAAP ENGINE

JavaScript state engine for claims, decisions, contradictions and source lineage. Explicit resolution records what changed and why; **31 core tests pass locally**. Explore implementation

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